Bias–Variance Tradeoff

Feature Combination and Model Complexity

Lecture Overview

In this lecture we study:

- Model complexity
- Feature combinations
- Polynomial features
- Interaction terms
- Nonlinear transformations
- Impact on bias and variance

Application context:
Cybersecurity detection systems

Motivation

Many real-world problems are not linear.
Example:
Phishing detection.
Features:

- URL length
- number of special characters
- domain age

Simple linear model:

$$y = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

But real patterns may be more complex.

Example Security Rule

Suppose phishing occurs when:

- URL is very long
- AND domain is very new

This means:

$$Risk \approx x_1 \times x_2$$

A simple linear model cannot capture this interaction.

Feature Engineering

Feature engineering creates new features from existing ones.
Examples:

- polynomial terms
- interaction terms
- nonlinear transformations

Goal:

Idea

Increase model expressiveness.

Polynomial Features

Given feature:

$$x$$

Polynomial expansion:

$$x^2, x^3, x^4$$

New model:

$$y = w_0 + w_1 x + w_2 x^2 + w_3 x^3$$

This allows curved decision boundaries.

Example: Network Traffic

Feature:

$$x = \text{packet rate}$$

Attack probability may increase rapidly:

$$y \approx x^2$$

Linear models cannot capture this behavior.
Polynomial features solve this.

Polynomial Model

General polynomial regression:

$$y = w_0 + w_1 x + w_2 x^2 + \cdots + w_d x^d$$

Where:

- d = polynomial degree
- larger d = more complex model

Effect on Model Complexity

Increasing polynomial degree increases complexity.
Example:

- Degree 1 $\rightarrow$ linear model
- Degree 2 $\rightarrow$ quadratic model
- Degree 5 $\rightarrow$ very flexible model

High flexibility may lead to overfitting.

Interaction Terms

Interaction terms combine multiple features.
Example:

$$x_1 = \text{URL length}$$

$$x_2 = \text{number of redirects}$$

Interaction feature:

$$x_1 x_2$$

Model with Interaction Terms

New regression model:

$$y = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1 x_2$$

Interpretation:
The effect of one feature depends on the other.

Cybersecurity Example

Intrusion detection features:

- connection duration
- number of packets

Attack signal may be strong when:

$$duration \times packets$$

is large.

Nonlinear Transformations

Sometimes we apply transformations:

- logarithm
- square root
- exponential

Example:

$$x' = \log(x)$$

Example: Malware File Size

Feature:

- file size.

Distribution may be skewed.
Better representation:

$$x' = \log(\text{file size})$$

This helps models learn patterns.

Feature Expansion

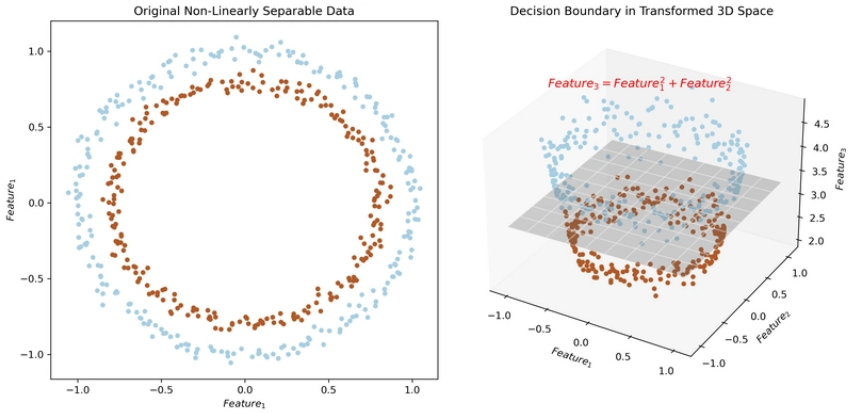
Suppose we have two features:

$$x_1, x_2$$

Polynomial expansion (degree 2):

$$1, x_1, x_2, x_1^2, x_2^2, x_1 x_2$$

Feature space becomes larger.



High Dimensional Feature Space

With many features:
Feature expansion grows quickly.
Example:

- 10 original features
- polynomial degree 3

New feature count becomes very large.
This increases model complexity.

Bias–Variance Tradeoff

Recall expected error:

$$Error = Bias^2 + Variance + Noise$$

Feature combinations influence both:

- decrease bias
- increase variance

Effect on Bias

Simple models have high bias.
Example:

- Linear model cannot capture nonlinear patterns.
- Feature combinations reduce bias by increasing flexibility.

Effect on Variance

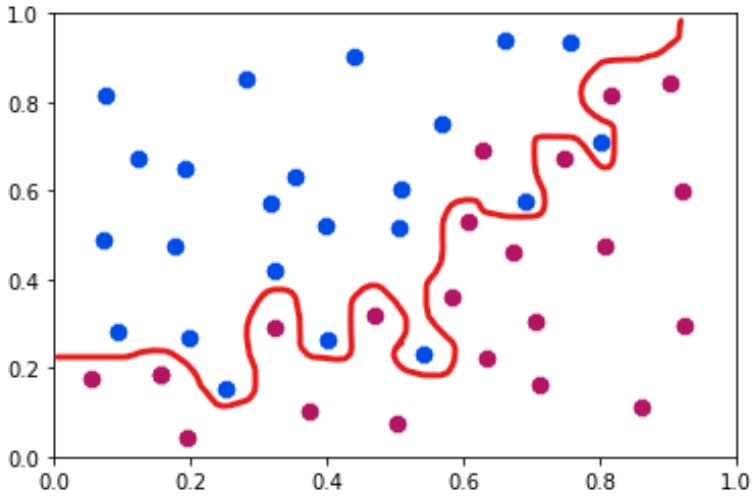
Complex models fit training data closely.
This increases variance.
Model becomes sensitive to noise.
Example:
memorizing attack patterns instead of learning general behavior.

Overfitting Example

Suppose we use:

$$degree = 10$$

Model perfectly fits training data.
But on new traffic data:
accuracy drops significantly.
This is overfitting.



Choosing Feature Complexity

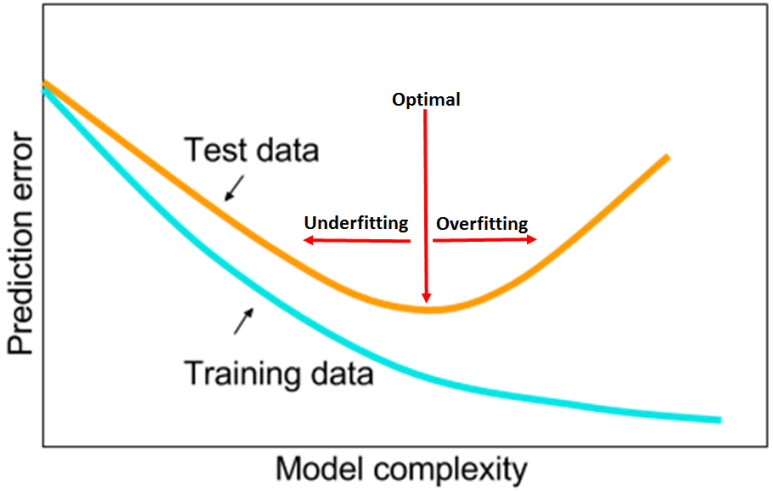
We must choose appropriate complexity.

Too simple:

- high bias
- poor detection

Too complex:

- high variance
- unstable model



Visual Interpretation

Low degree polynomial:

- underfits data.

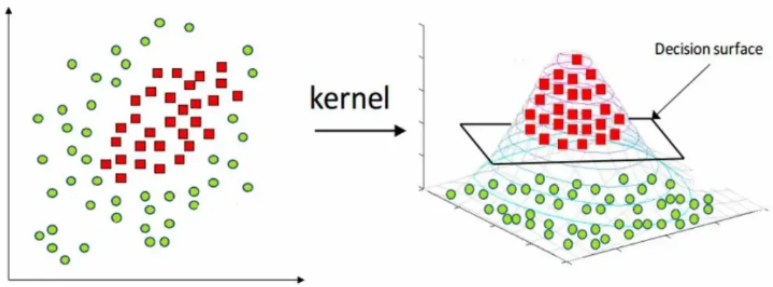
Medium degree:

- captures pattern.

High degree:

- overfits noise.

Feature combinations directly control this behavior.



Cybersecurity Detection Pipeline

Example system:

1. collect network logs
2. extract features
3. create feature combinations
4. train ML model
5. detect suspicious activity

Example Feature Set

Phishing detection features:

- URL length
- number of dots
- domain age
- number of redirects

Engineered features:

- URL length $\times$ redirects
- log(domain age)
- redirects^2

Advantages of Feature Combination

Benefits:

- capture nonlinear patterns
- improve predictive power
- increase model flexibility

Risks of Feature Combination

Potential problems:

- high dimensionality
- overfitting
- increased computation

Therefore we must control complexity.

Regularization Connection

Regularization helps control complex feature spaces.
Example Ridge regression:

$$\min \sum (y_i - \hat{y}_i)^2 + \lambda \sum w_j^2$$

This prevents large coefficients.

Practical Workflow

Typical ML pipeline:

1. collect cybersecurity dataset
2. engineer feature combinations
3. train model
4. control complexity using regularization
5. evaluate model

Key Takeaways

- Feature combinations increase model complexity
- Polynomial features capture nonlinear patterns
- Interaction terms capture feature relationships
- Complexity affects bias and variance